Central-Potential Scattering, the Rutherford Cross Section, and Hamiltonian Mechanics

HC 6001 — Classical Mechanics

October 22, 2024

1 Scattering by a central potential — setup

Definition 1 (Scattering by a central potential)**.** *Consider a beam of particles scattered by a central potential $V(r)$. The incident energy and velocity are*

$$E = \frac{m}{2}v_0^2, \qquad v_0 = \sqrt{\frac{2E}{m}}. \tag{1}$$

The orbital angular momentum is fixed by the impact parameter s,

$$l = msv_0 = s\sqrt{2mE}, \qquad s = \frac{l}{\sqrt{2mE}}. \tag{2}$$

In spherical coordinates the volume element is $d^3r = r^2\,dr\,d\Omega$ with $d\Omega = \sin\theta\,d\theta\,d\phi$. Because the azimuthal angle ϕ is irrelevant by symmetry, the solid-angle element for scattering into deflection angle θ is

$$d\Omega = 2\pi\sin\theta\,d\theta. \tag{3}$$

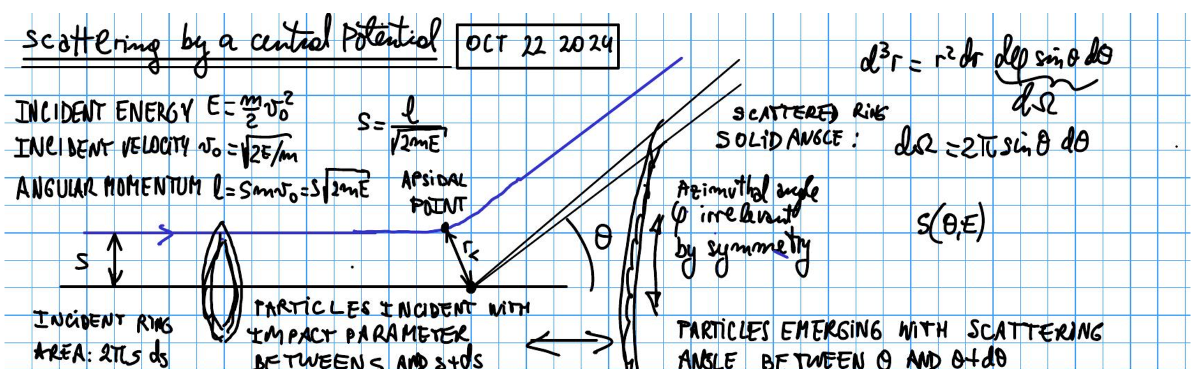

Figure 1: Scattering trajectory: impact parameter s, apsidal point at r_c, and deflection angle θ; particles incident in the ring between s and $s + ds$ emerge between θ and $\theta + d\theta$.

2 Incident intensity and differential cross section

Let the incident intensity I be the number of particles incident per unit time per unit beam cross-sectional area. The differential cross section $\sigma(\Omega)$ is defined by

$$\sigma(\Omega)\,d\Omega = \frac{\#\text{ particles scattered into } d\Omega \text{ per unit time}}{I}. \tag{4}$$

Particles incident through the ring between impact parameters s and $s + ds$ occupy area $2\pi s\,|ds|$ and emerge with deflection angle between θ and $\theta + d\theta$. Conservation of particle number gives

$$I\,(2\pi s)\,|ds| = I\,\sigma(\theta)\,2\pi\sin\theta\,|d\theta|, \tag{5}$$

so that

$$\sigma(\theta) = \frac{s}{\sin\theta}\left|\frac{ds}{d\theta}\right|. \tag{6}$$

The total cross section is

$$\sigma_T = \int d\Omega\,\sigma(\Omega). \tag{7}$$

3 Repulsive vs. attractive scattering geometry

Let ψ be the angle between the incident asymptote (the horizontal) and the apsidal direction. The geometry relates ψ to the deflection angle θ differently in the two cases:

$$\text{repulsive:}\quad \theta = \pi - 2\psi, \qquad \psi = \frac{\pi - \theta}{2}, \tag{8}$$

$$\text{attractive:}\quad \theta = 2\psi - \pi, \qquad \psi = \frac{\pi + \theta}{2}, \tag{9}$$

and in all cases

$$\theta = |\,\pi - 2\psi\,|. \tag{10}$$

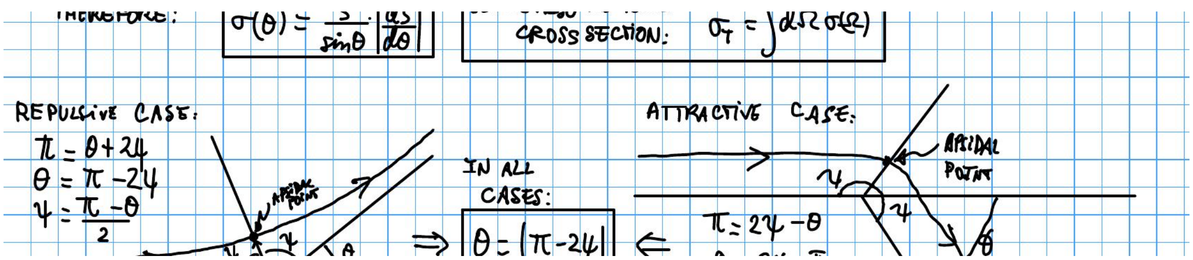

Figure 2: Repulsive (left) and attractive (right) central-potential trajectories, showing the apsidal point and the angle ψ to the horizontal.

4 Universal scattering-angle integral for any $V(r)$

To obtain $s(\theta, E)$ for an arbitrary central potential, begin from energy conservation including the centrifugal term,

$$E = \frac{m}{2}\dot{r}^2 + V(r) + \frac{l^2}{2mr^2}. \tag{11}$$

Solving for the radial velocity,

$$\dot{r} = \left[\frac{2}{m}\left(E - V(r) - \frac{l^2}{2mr^2}\right)\right]^{1/2}, \qquad dt = \frac{dr}{\dot{r}}. \tag{12}$$

With $l = mr^2\dot{\theta}$ we have $\dot{\theta} = l/(mr^2)$, hence

$$d\theta = \dot{\theta}\,dt = \frac{l}{mr^2}\,dt = \frac{l\,r^{-2}\,dr}{\sqrt{2m\left(E - V(r) - \dfrac{l^2}{2mr^2}\right)}}. \tag{13}$$

Substituting $l = s\sqrt{2mE}$ and dividing through by E,

$$d\theta = \frac{s\,r^{-2}\,dr}{\sqrt{1 - \dfrac{V(r)}{E} - \dfrac{s^2}{r^2}}}. \tag{14}$$

Integrating from the distance of closest approach r_m to infinity gives the apsidal angle

$$\psi = \int_{r_m}^{\infty} \frac{s\,r^{-2}\,dr}{\sqrt{1 - \dfrac{V(r)}{E} - \dfrac{s^2}{r^2}}}. \tag{15}$$

With the substitution $u = 1/r$, $du = -r^{-2}\,dr$, and $u_m = 1/r_m$,

$$\psi = \int_0^{u_m} \frac{s\,du}{\sqrt{1 - \dfrac{V(u)}{E} - s^2u^2}}. \tag{16}$$

Using $\theta = |\pi - 2\psi|$, the scattering angle for any central potential is

$$\theta(s) = \left| \pi - 2\int_0^{u_m} \frac{s\,du}{\sqrt{1 - \dfrac{V(u)}{E} - s^2u^2}} \right|. \tag{17}$$

From $\theta(s)$ one obtains $|ds/d\theta| = 1/|d\theta/ds|$ and hence $\sigma(\theta)$ through (6). **Up to this point every result is valid for any central potential $V(r)$.**

5 Coulomb scattering — hyperbolic orbits and eccentricity

Definition 2 (Coulomb scattering)**.** *For the Coulomb interaction the force and potential are*

$$f = \frac{ZZ'e^2}{r^2}, \qquad V(r) = -\frac{k}{r}, \qquad k = -ZZ'e^2. \tag{18}$$

For $V = -k/r$ only*, the orbit type is set by the sign of the energy:*

- $E > 0$*: hyperbolic orbit (scattering),*
- $E = 0$*: parabolic orbit,*
- $E < 0$*: elliptic orbit.*

For scattering ($E > 0$) the orbit equation is

$$\frac{1}{r} = C\big[1 + \epsilon\cos(\theta - \theta')\big], \tag{19}$$

with coefficient and eccentricity

$$C = \frac{mk}{l^2}, \qquad \epsilon = \left[1 + \frac{2El^2}{mk^2}\right]^{1/2} = \left[1 + \left(\frac{2sE}{ZZ'e^2}\right)^2\right]^{1/2} > 1, \tag{20}$$

where $l = s\sqrt{2mE}$ was used. Two physically distinct situations arise:

i) **repulsive**: $ZZ' > 0 \Rightarrow k = -ZZ'e^2 < 0$;

ii) **attractive**: $ZZ' < 0 \Rightarrow k > 0$ (also gravity, $k = Gm_1m_2$).

6 Coulomb scattering — repulsive case

Case i (repulsive): here $k < 0$, so $C = mk/l^2 < 0$. Positivity of $1/r$ requires

$$1 + \epsilon\cos(\theta - \theta') < 0 \quad \Longleftrightarrow \quad \cos(\theta - \theta') < -\frac{1}{\epsilon}, \tag{21}$$

so $\theta - \theta' = 0$ lies inside the inaccessible region and periapsis occurs at $\theta' = \pi$. Using $\cos(\theta - \pi) = -\cos\theta$,

$$\frac{1}{r} = \frac{mZZ'e^2}{l^2}\big(\epsilon\cos\theta - 1\big), \tag{22}$$

which is maximal (radius minimal) at the apsidal point $\theta = 0$,

$$\frac{1}{r_m} = \frac{mZZ'e^2}{l^2}\,(\epsilon - 1). \tag{23}$$

As $r \to \infty$ the bracket vanishes, giving $\cos\psi = 1/\epsilon$. With $\psi = (\pi - \theta)/2$,

$$\frac{1}{\epsilon} = \cos\frac{\pi - \theta}{2} = \sin\frac{\theta}{2}. \tag{24}$$

7 Coulomb scattering — attractive case

Case ii (attractive): here $k > 0$, so $C = km/l^2 > 0$. Take $\theta' = 0$, so

$$\frac{1}{r} = C\big[1 + \epsilon\cos\theta\big], \tag{25}$$

maximal at the apsidal point $\theta = 0$. As $r \to \infty$, $\cos\psi = -1/\epsilon$. With $\psi = (\pi + \theta)/2$,

$$-\frac{1}{\epsilon} = \cos\frac{\pi + \theta}{2} = -\sin\frac{\theta}{2} \quad \Longrightarrow \quad \frac{1}{\epsilon} = \sin\frac{\theta}{2}, \tag{26}$$

identical to the repulsive relation.

8 Rutherford differential cross section

Combining both cases through $1/\epsilon = \sin(\theta/2)$,

$$\cot^2\frac{\theta}{2} = \frac{1}{\sin^2\frac{\theta}{2}} - 1 = \epsilon^2 - 1 = \left(\frac{2sE}{ZZ'e^2}\right)^2, \tag{27}$$

so the impact parameter as a function of angle is

$$s(\theta, E) = \left|\frac{ZZ'e^2}{2E}\right| \cot\frac{\theta}{2}. \tag{28}$$

Differentiating,

$$\frac{ds}{d\theta} = -\left|\frac{ZZ'e^2}{2E}\right| \frac{1}{2\sin^2\frac{\theta}{2}} = -\left|\frac{ZZ'e^2}{4E}\right| \frac{1}{\sin^2\frac{\theta}{2}}, \tag{29}$$

and substituting into (6) with $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$,

$$\sigma(\theta) = \frac{1}{4}\left(\frac{ZZ'e^2}{2E}\right)^2 \csc^4\frac{\theta}{2}. \tag{30}$$

This is the **Rutherford scattering cross section** for the Coulomb force.

Remark 1 (Properties of the Rutherford cross section)**.** *1. It is the same for the repulsive and attractive cases, since it depends only on $(ZZ')^2$.*

2. It diverges like E^{-2} at low incident energies.

3. It diverges like θ^{-4} (very strongly) at small scattering angles.

4. The classical result is exactly the same as the quantum-mechanical result.

The total cross section is

$$\sigma_T = \int \sigma(\Omega)\, d\Omega, \qquad \sigma(\theta) = \text{const} \times \frac{1}{\sin^4\frac{\theta}{2}}. \tag{31}$$

Writing $d\Omega = 2\pi \sin\theta\, d\theta = 4\pi \sin\frac{\theta}{2}\cos\frac{\theta}{2}\, d\theta$,

$$\sigma_T = \frac{4\pi}{4}\left(\frac{ZZ'e^2}{2E}\right)^2 \int_0^\pi \frac{\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{\sin^4\frac{\theta}{2}}\, d\theta \;\sim\; \int_0 \frac{d\theta}{(\theta/2)^3} = 8\int_0 \frac{d\theta}{\theta^3}, \tag{32}$$

whose lower limit behaves like $\theta^{-2}\big|_0 \to +\infty$. Hence σ_T is **strongly divergent**.

9 Notation, divergent total cross section, screened potentials

Remark 2 (Notation and long-range potentials)**.** *Goldstein writes $\sigma(\Omega)$ and σ_T for the differential and total cross sections, where other texts write $d\sigma/d\Omega$ and σ, respectively.*

Classically, $\sigma_T = +\infty$ for any *long-range potential. Quantum mechanically, $\sigma_T = +\infty$ only for some long-range potentials $V(r)$ that do not decay fast enough as $r \to \infty$.*

Caveat: *in a dense conducting material the Coulomb potential becomes a screened (Yukawa) potential,*

$$V_{\text{screen}}(r) = -\frac{k}{r}\, e^{-r/\lambda}. \tag{33}$$

In non-conducting materials the screened potential may take a different form but still vanishes much faster than $1/r$, rendering σ_T finite.

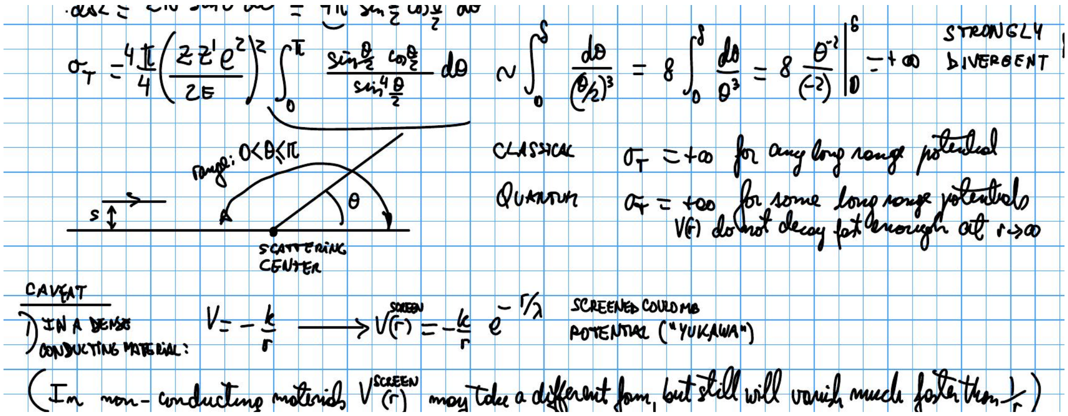

Figure 3: Scattering-center geometry, notation comparison, and the screened Coulomb (Yukawa) caveat (from the lecture notes).

10 Hamiltonian mechanics — Lagrangian review

Definition 3 (Lagrangian formalism)**.** *The Lagrangian is*

$$L = T - V = L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n, t), \tag{34}$$

and the Euler–Lagrange equations of motion are

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = 0, \tag{35}$$

with generalized (canonical) momentum

$$p_i = \frac{\partial L}{\partial \dot{q}_i}. \tag{36}$$

Remark 3 (Midterm remark)**.** *In polar coordinates,* $\dfrac{\partial L}{\partial \dot{\theta}} = p_\theta = l_z$.

11 Legendre transformation (math and thermodynamics)

We wish to change variables from $(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n)$ to $(q_1, \ldots, q_n, p_1, \ldots, p_n)$. For a function $f(x, y)$,

$$df = \frac{\partial f}{\partial x}\, dx + \frac{\partial f}{\partial y}\, dy. \tag{37}$$

Thermodynamics example (fluid at fixed particle number N). The first law is

$$dU = \delta Q + \delta W = T\, dS - p\, dV, \qquad U = U(S, V). \tag{38}$$

To use (T, V) as variables, define the Helmholtz free energy $A = U - TS$; then

$$dA = dU - d(TS) = T\, dS - p\, dV - T\, dS - S\, dT = -S\, dT - p\, dV, \tag{39}$$

so $A = A(T, V)$.

General Legendre transform. For $f(x, y) = u\, dx + v\, dy$ with $u = \partial f/\partial x$ and $v = \partial f/\partial y$, transform $(x, y) \to (u, y)$ by defining $g = f - ux$. Then

$$dg = df - d(ux) = u\, dx + v\, dy - u\, dx - x\, du = -x\, du + v\, dy, \tag{40}$$

so $g = g(u, y)$.

12 Hamiltonian and canonical equations of motion

Apply the Legendre transform to $L(q_1, \ldots, q_n, \dot{q}_1, \ldots, \dot{q}_n)$ with $p_i = \partial L/\partial \dot{q}_i$, defining the Hamiltonian

$$H = -\Big(L - \sum_i p_i \dot{q}_i\Big) = \sum_i p_i \dot{q}_i - L. \tag{41}$$

Taking the differential,

$$dH = -dL + \sum_i d(p_i \dot{q}_i) = -\frac{\partial L}{\partial t}\, dt + \sum_i \left(\dot{q}_i\, dp_i - \frac{\partial L}{\partial q_i}\, dq_i\right), \tag{42}$$

so that $H = H(p_1, \ldots, p_n, q_1, \ldots, q_n, t)$.

Theorem 1 (Canonical equations of motion)**.** *Reading off the coefficients of dp_i, dq_i, and dt in dH, and using the Lagrange equation $\partial L/\partial q_i = \dot{p}_i$,*

$$\frac{\partial H}{\partial p_i} = \dot{q}_i, \tag{43}$$

$$\frac{\partial H}{\partial q_i} = -\frac{\partial L}{\partial q_i} = -\dot{p}_i, \tag{44}$$

$$\frac{\partial H}{\partial t} = -\frac{\partial L}{\partial t}. \tag{45}$$

These are the ***canonical equations of motion****.*

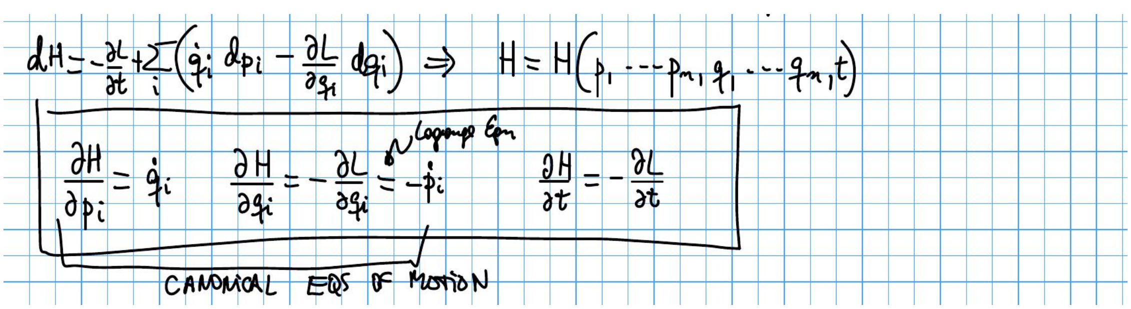

Figure 4: Boxed canonical equations of motion (from the lecture notes).